

Relational model — attributes.
key.

Relational algebra:

σ select operator, select row

$\wedge$ and

$\vee$ or

$\neg$ not

example $\sigma_{dept-name = "physics"}(instructor)$
relation.

π select columns. (31)

example: $\pi_{ID, name}(instructor)$

Cartesian — product.

example. $instructor \times teaches,$

4

5

不多条件就是

(4x5)

Natural join.

example instructor $\bowtie$ teaches
need common attributes.

Theta join.

$$Exp_1 \bowtie_{\theta} Exp_2 = \sigma_{\theta} (Exp_1 \times Exp_2)$$

Wrap up

- Relational algebra
 - Simplest query: relation name
 - Use operators to filter, slice, combine
 - Operators so far: select, project, cross-product, natural join, theta join

Feature	Theta Join (θ -Join)	Natural Join ($\bowtie$)
Condition	Any comparison (e.g., =, >, <, !=)	Only equality (=)
Duplicate Columns?	No removal (keeps all)	Removes duplicates
Symbol	$R \bowtie_{\theta} S$	$R \bowtie S$

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Union " $\cup$ " 交集

$\pi_{name} (instructor) \cup \pi_{dept-name} (department)$

intersection " $\cap$ " 子集

Set difference.

" - "

去除

$\Pi_{\text{dept-name}}(\text{department}) - \Pi_{\text{dept-name}}(\text{instructor})$

Rename

" ρ "

$\rho_{X(A)}(E)$

→ 修改的 relation name.

new relation name

new attributes name.

example.

$\rho_{r_1(a_1, b_1, c_1, d_1)} \text{Instructor} \bowtie \rho_{r_2(a_2, b_2, c_2, d_2)} \text{Instructor}$

$\sigma_{r_1.c_1 = r_2.c_2}$

Assignment Operation.

" $\leftarrow$ "

DDL (Data Definition Language)

DML (Data manipulation language)
操作.

Basic query structure.

select $A_1, A_2 \dots$ from $r_1, r_2 \dots$ where P

Update.

insert into r_i value

delete from r_i

drop table r_i

alter

numeric, char...

alter table r_i add A_i 格式

alter table r_i drop A_i

distinct 去重

*

as 重命名

in

between

%

match any substring

任意长度

_

match any character

一位长度

avg

min

max

sum

count.

Subquery:

Some. 其中一个

all

with.